The following rules therefore govern:

When forward motion is taken from the *lower half* of the link.

In *forward motion*: to move the valve ahead, lengthen the eccentric rod. To move it back, shorten the eccentric rod.

In *backward motion*: to move the valve ahead, shorten the eccentric rod. To move it back, lengthen the eccentric rod.

When forward motion is taken from the *upper half* of the link.

In *forward motion*: to move the valve back, lengthen the eccentric rod. To move it ahead, shorten the eccentric rod.

In *backward motion*: to move the valve ahead, lengthen the eccentric rod. To move it back, shorten the eccentric rod.

A change in the length of the eccentric rod will not alter the position of the valve a like amount. The two will have approximately the same ratio to each other as that between the eccentric crank throw and the valve travel.

For example: suppose the eccentric throw is 12 inches and the valve travel 6 inches. The ratio is two to one or the latter is one-half of the former. In such a case, a change in the length of the eccentric rod of $\frac{1}{8}$ inch would alter the position of the valve only one-half of that amount, or $\frac{1}{16}$ inch.

The following method of setting the Walschaert Valve gear is equally applicable to designs having constant lead and those having variable lead.

*First*: Check the lengths of the eccentric crank, the lap and lead lever, and the lap and lead lever connector.

These dimensions should conform to the drawings and should not be altered.

*Second*: Raise the main wheels so that the distance from the center of the wheel to the top of the frame conforms to the amount specified, minus $\frac{1}{2}$ inch for wear of brasses. Then find the dead centers and port marks in the usual manner.

*Third*: Assemble the gear complete, temporarily tightening the eccentric crank in a position which will give the specified throw.

*Fourth*: Place the radius bar in the center of the link. Mark the mid-gear position. Then get the lead at each end of the cylinder. When the lead is constant, the *average* lead, or the sum of the leads on opposite ends divided by two, should be equal to the specified lead in full gear. When the lead is variable, the *average* lead in mid-gear position should be equal to one-half of the sum of the specified leads in full forward and full back gear. In other words, it should be the lead due to the lap and lead lever, unaffected by the position of the eccentric crank. Any error in the average lead when the radius bar is in central position is due to an error in the length of the upper or lower arms of the lap and lead lever.

Having thus checked the lengths of the lap and lead lever, equalize the lead by means of the adjusting nuts on the valve stem or by changing the length of the radius bar.

*Fifth*: drop the lever into forward gear until the specified travel is obtained. Then, if the average lead is equal to the specified lead in full gear, the eccentric crank is correctly set. If this is not the case, the eccentric crank should be driven one way or the other until the error is corrected. If the average lead is less than the specified lead, the eccentric crank should be driven *inward*, if it *leads* the main pin; and driven *outward*, if it *follows* the main pin. If the average lead is more than the specified lead, the eccentric crank should be driven *outward*, if it *leads* the main pin; and driven *inward* if it *follows* the main pin.

After eccentric crank is correctly set, check valve travel; and relocate full forward position of the reverse lever.

*Sixth*: If the average lead is correct but unequally divided on the front and back centers, lengthen or shorten the eccentric rod, according to the rules given on this page until it is equalized. It must be borne in mind that to change the lead a given amount the eccentric rod must be changed a greater amount, or about in proportion as the eccentric throw is greater than the valve travel. (See figure 26).

Place the reverse lever in a position that will give full travel to back gear, marking this position on the quadrant; and check the lead in the same manner. With variable lead, the full back gear lead should be as much greater than the lead at mid gear as the lead at mid gear is greater than that at full forward gear.

*Seventh*: Run over the cut-offs and obtain other events for as many positions as required. In running over the cut-offs of locomotives of the articulated type, obtain the cut-offs for each position of the lever for *both engines* before moving the lever to a new position. This is necessary in order that the relative cut-offs in high and low pressure cylinders may be compared.

*Note*: Do not attempt to square cut-off at the expense of lead and port opening.

## VARIABLE LEAD

When this setting is resorted to, it is advisable to adjust the gear according to the general rule, using a temporarily fixed eccentric crank to obtain the proper length of the eccentric rod and valve location, afterwards, readjust the eccentric crank to suit the desired lead in full gear.